

DR. BABASAHEB AMBEDKAR MARATHWADA UNIVERSITY, CHHATRAPATI
SAMBHAJINAGAR (M.S.)

MAJOR PROJECT REPORT ON: IOT BASED COTTON QUALITY MONITORING SYSTEM

Submitted by

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Imran Patel

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Aadil Shaikh

Under the guidance of

Prof. S. B. Biradar

**In partial fulfillment of the award of
Bachelor of Technology (Electronics
& Computer Engineering)**



**Department of Electronics & Computer Engineering
Maharashtra Institute of Technology,
(An Autonomous Institute)**

Chhatrapati Sambhajnagar, Maharashtra – 431005

2025-26

CERTIFICATE

This is to certify that the Major Project final report entitled “**IOT Based Cotton Quality Monitoring System**”, submitted by **Zaid Khan, Imran Patel, Rupesh Patil and Aadil Shaikh** are the Bonafide work completed under my supervision and guidance in partial fulfillment for the award of Bachelor of Technology (**Electronics & Computer Engineering**) of Dr. Babasaheb Ambedkar Marathwada University, Chhatrapati Sambhajanagar.

Place: Chh. Sambhajanagar (Aurangabad)

Date: / /2025

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ACKNOWLEDGEMENT

The portion of success is brewed by the efforts put in by many individuals. It is constant support provided by people who give you the initiative, who inspire you at each step of your endeavor that eventually helps you in your goal.

I wish to express my deep gratitude and heartily appreciation for the invaluable guidance of our professors throughout the span of doing this minor project. We are indebted to our college **Director Dr. N. G. Patil**.

I am also thankful to our **HOD Dr. S. J. Nandedkar** and my **Project Guide Prof. S. B. Biradar** for his invaluable and elaborate suggestions. Their excellent guidance helped me to complete this task successfully.

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ABSTRACT

The IoT-based cotton Quality monitoring system is designed to enhance the quality assessment of cotton by continuously monitoring key parameters during its production. It utilizes various sensors to measure important factors such as Staple Length, moisture content, color grade, R.D. (Reflectance-Degree) value and Trash Content. The staple length is the average length of a group of fibers that make up cotton yarn. The moisture sensors help in determining the optimal water content in the cotton, preventing issues like over- drying or under-drying that can impact its quality. Color sensors assess the color of cotton, which is crucial in determining its grade and market value. The R D value is used to evaluate the cotton's cleanliness and degree of contamination through reflectance measurement. Additionally, the Trash Content in cotton refers to non-fiber particles such as leaf, seedcoat, bark, grass, dust and other foreign matters. By collecting data of these parameters, the system enables better decision-making, ensures consistent cotton quality, and enhances overall efficiency in cotton processing, allowing for timely interventions and optimized management throughout the production cycle.

1. INTRODUCTION

1.1 Introduction

In the modern textile industry, cotton remains one of the most widely used natural fibers due to its versatility, comfort, and durability. However, the quality of cotton can vary significantly depending on factors such as Staple length, moisture content, and contamination levels. Ensuring high-quality cotton is crucial for producing superior textile products, meeting industry standards, and satisfying consumer demands. Traditional methods of cotton quality assessment often rely on manual inspection and laboratory testing, which can be time-consuming, labor-intensive, and prone to human error.



To address these challenges, the development of an automated Cotton Quality Monitoring System has become essential. This system leverages advanced technologies such as IoT (Internet of Things), machine learning, and computer vision to provide monitoring and analysis of cotton quality parameters. By automating the quality assessment process, the system aims to improve accuracy, reduce production costs, and enhance overall efficiency in the textile supply chain.

1.2 Need of Topic

The need for an IoT-based cotton monitoring system arises from the importance of ensuring high-quality cotton production. Cotton is a vital raw material for the textile industry, and its quality directly affects the final products made from it. Monitoring key factors like Staple Length, moisture, color grade, and R.D value helps in maintaining consistent quality. Without proper monitoring, cotton can be over-dried or under-dried, leading to poor quality, or it might not meet the required color or fiber quality standards. The IoT system helps by providing, allowing farmers and producers to make quick decisions and take necessary actions. This results in better quality cotton, higher efficiency in processing, and reduced waste, which ultimately benefits the entire cotton production and textile industry.

1.3 Objective of Topic

1. Automated Quality Assessment – Develop a system that automatically detects and classifies cotton quality based on parameters like Staple length, strength, color, Trash and moisture content.
2. Standardization & Accuracy – Ensure compliance with industry standards by integrating precision sensors and image processing for accurate quality detection.
3. Energy – efficient and Scalable System – Design a low power , scalable IOT framework that can be implemented across different cotton producing regions.
4. Cost- effectiveness and Efficiency – Reduce manual labor and subjectivity in cotton grading while improving efficiency in cotton processing industries

1.4 Cotton Production

Cotton is grown in many countries, including India, China, the United States, and Brazil. India is the world's largest producer of cotton, followed by China. In 2023/2024, India produced 25.4 million bales of cotton, while China produced 27.35 million bales.

Table 1.4.1 Production of Cotton – Global Scenario for last 5 years

Countries	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23
World	27.00	25.98	26.26	23.99	25.18	24.51
India	6.29	5.66	6.21	5.99	5.29	5.84
China	5.89	6.04	5.80	5.91	5.73	5.98
USA	4.56	4.00	4.33	3.18	3.81	3.15
Brazil	2.00	2.78	3.00	2.36	2.55	3.02
Pakistan	1.80	1.67	1.46	0.96	1.27	0.85

In India, cotton is harvested in December–January in the north and January–May in the south.

The north zone states are Punjab, Haryana, and Rajasthan, and the south zone states are Andhra Pradesh, Telangana, Karnataka, and Tamil Nadu are the major cotton-producing states in India.

In India, the leading cotton-producing states are Maharashtra, Gujarat, Andhra Pradesh, and Madhya Pradesh. In Which Gujarat Is a Largest Cotton Producing State In India.

Table 1.4.2 State wise Cotton Production
(AREA IN LAKH HECTARE, PROD IN LAKH BALES 170 KGS, YIELD KGS PER HECT)

Year	2018-19			2019-20			2020-21			2021-22			2022-23(P)*			2023-24(P)*		
State	Area	Prod	Yield	Area	Prod	Yield	Area	Prod	Yield	Area	Prod	Yield	Area	Prod	Yield	Area	Prod	Yield
Punjab	2.68	8.50	539	2.48	9.50	651	2.52	10.23	690	2.51	6.46	438	2.49	4.44	303	1.69	4.89	492
Haryana	7.08	23.00	552	7.23	26.50	623	7.40	18.23	419	6.36	13.16	352	5.75	10.01	296	6.83	14.50	361
Rajasthan	6.29	27.50	743	7.6	29.00	649	8.07	32.07	676	7.56	24.81	558	8.15	27.74	579	7.91	27.43	590
North total	16.05	59.00	625	17.31	65.00	638	17.99	60.53	572	16.43	44.43	460	16.39	42.19	438	16.43	46.82	484
Gujarat	26.60	90.00	575	26.55	89.00	570	22.70	72.18	541	22.84	75.09	559	24.84	87.95	602	26.83	89.65	568
Maharashtra	42.18	76.00	306	44.91	87.00	329	45.44	101.05	378	44.10	82.49	318	41.82	83.16	338	42.34	82.43	331
Madhya Pradesh	6.14	23.00	637	6.50	20.00	523	5.88	13.38	387	5.60	14.20	431	5.95	14.33	409	6.30	17.97	485
Central total	74.92	189.00	429	77.96	196.00	427	74.02	186.61	429	72.54	171.78	403	72.61	185.44	434	75.47	190.05	428
Telangana	18.39	42.00	388	21.27	54.00	432	23.58	57.97	418	18.89	48.78	439	19.73	57.45	495	18.18	48.12	450
Andhra Pradesh	6.20	15.00	411	6.57	18.00	466	6.06	16.00	449	5.54	17.08	524	7.04	15.41	372	4.27	11.58	461
Karnataka	7.18	16.00	379	8.17	20.00	416	8.20	23.20	481	6.74	19.55	493	9.49	25.68	460	6.95	16.93	414
Tamil Nadu	1.33	6.00	767	1.70	6.00	600	1.12	2.43	369	1.48	3.02	347	1.73	3.19	313	0.91	1.95	364
South Total	33.10	79.00	406	37.71	98.00	442	38.96	99.60	435	32.65	88.43	460	37.99	101.73	455	30.31	78.58	441
Orissa	1.57	4.00	433	1.7	4.00	400	1.71	5.51	548	1.93	6.26	551	2.16	7.05	555	2.36	7.44	536
Others	0.50	2.00	680	0.09	2.00	3778	0.17	0.23	230	0.16	0.27	287	0.12	0.19	269	0.12	0.22	312
GR TOTAL	126.14	333.00	449	134.77	365.00	460	132.85	352.48	451	123.71	311.17	428	129.27	336.60	443	124.69	323.11	441

Note: Production is inclusive of state-wise loose cotton production:-

- 1) from 2010-11 to 2018-19 : 26.10 lakh bales as per survey undertaken by ATIRA
- 2) from 2019-20 onwards : 5.45 lakh bales as per survey undertaken by SVPISTM

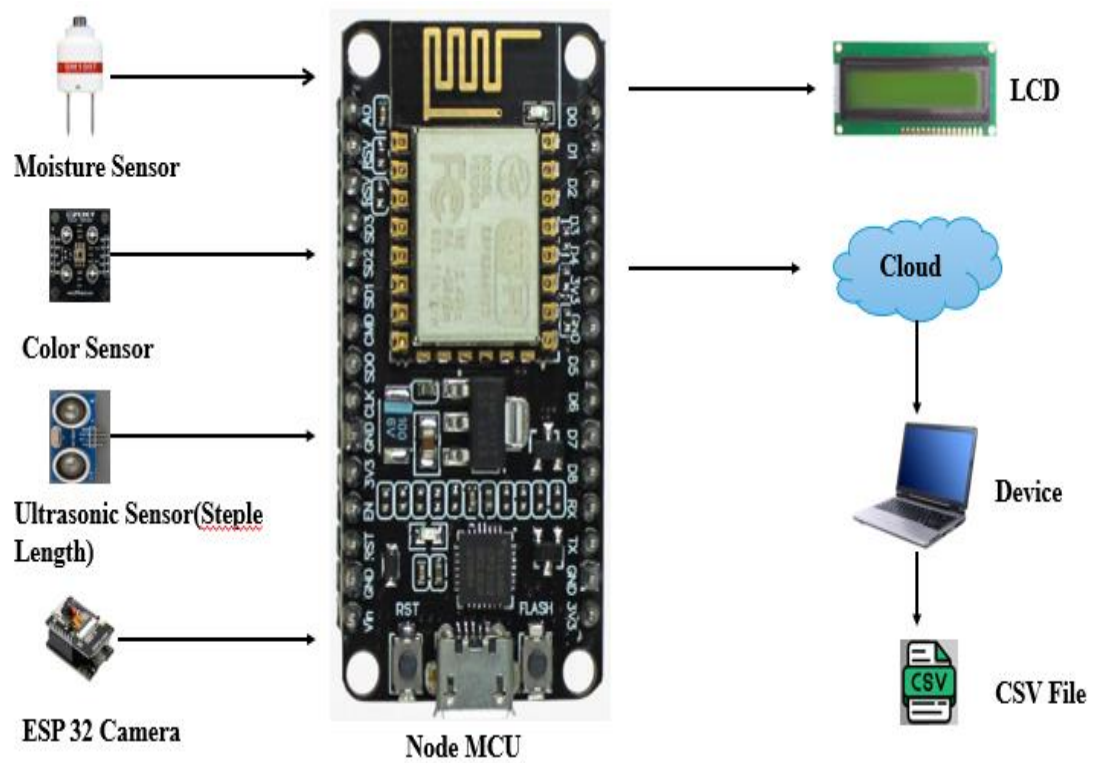
P-Provisional

Source: Meeting of Committee on Cotton Production and Consumption (COCPC) held on 14-03-2024

1.5 Cotton Quality

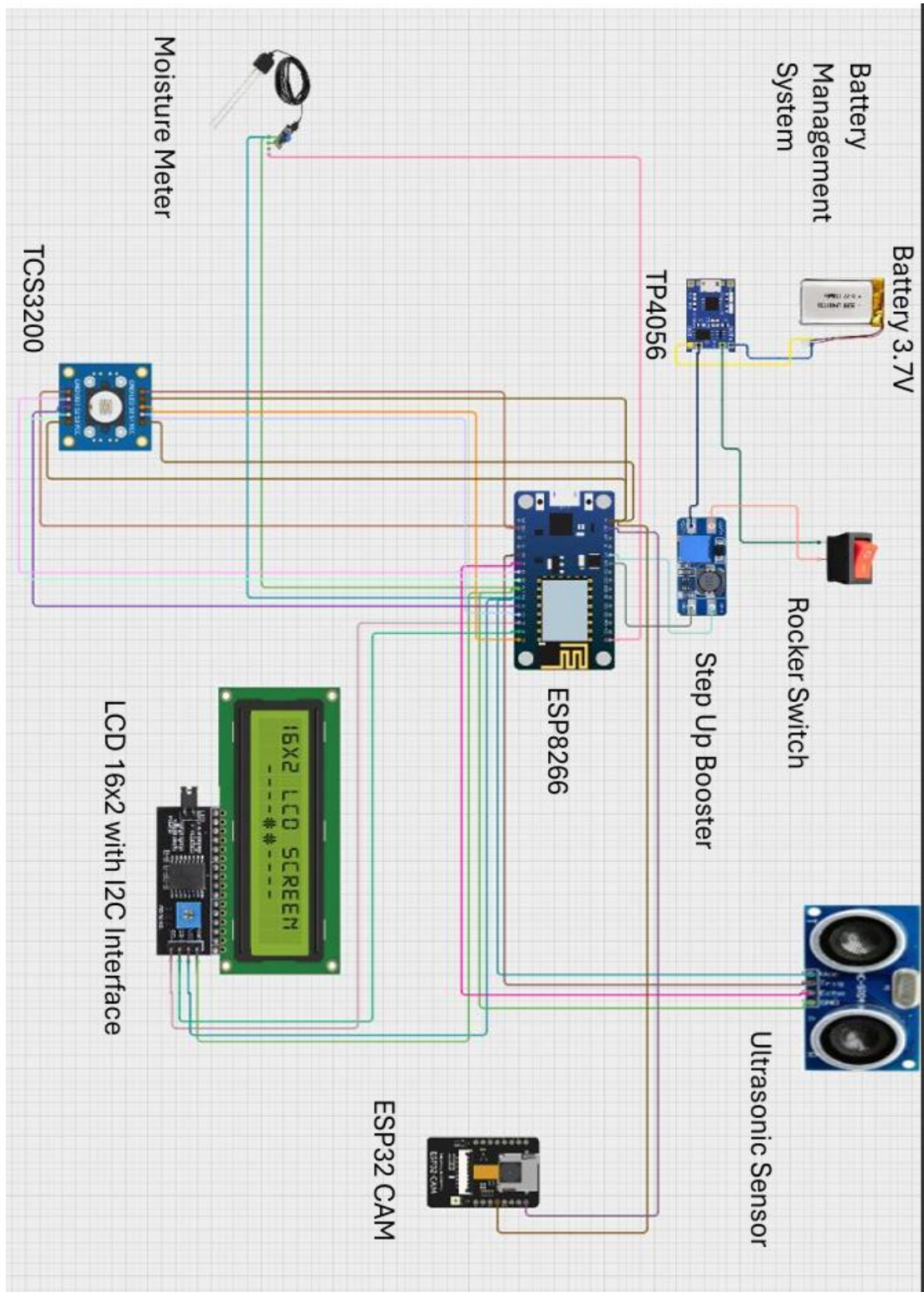
Quality refers to the degree of excellence or superiority of something, often measured against standards, expectations, or specifications. It can apply to products, services, processes, or experiences. In the context of cotton, quality refers to the characteristics of the cotton fiber that determine its suitability for various end uses, such as textiles, clothing, or industrial products. The quality of cotton is evaluated based on several key factors.

1.6 Block Diagram



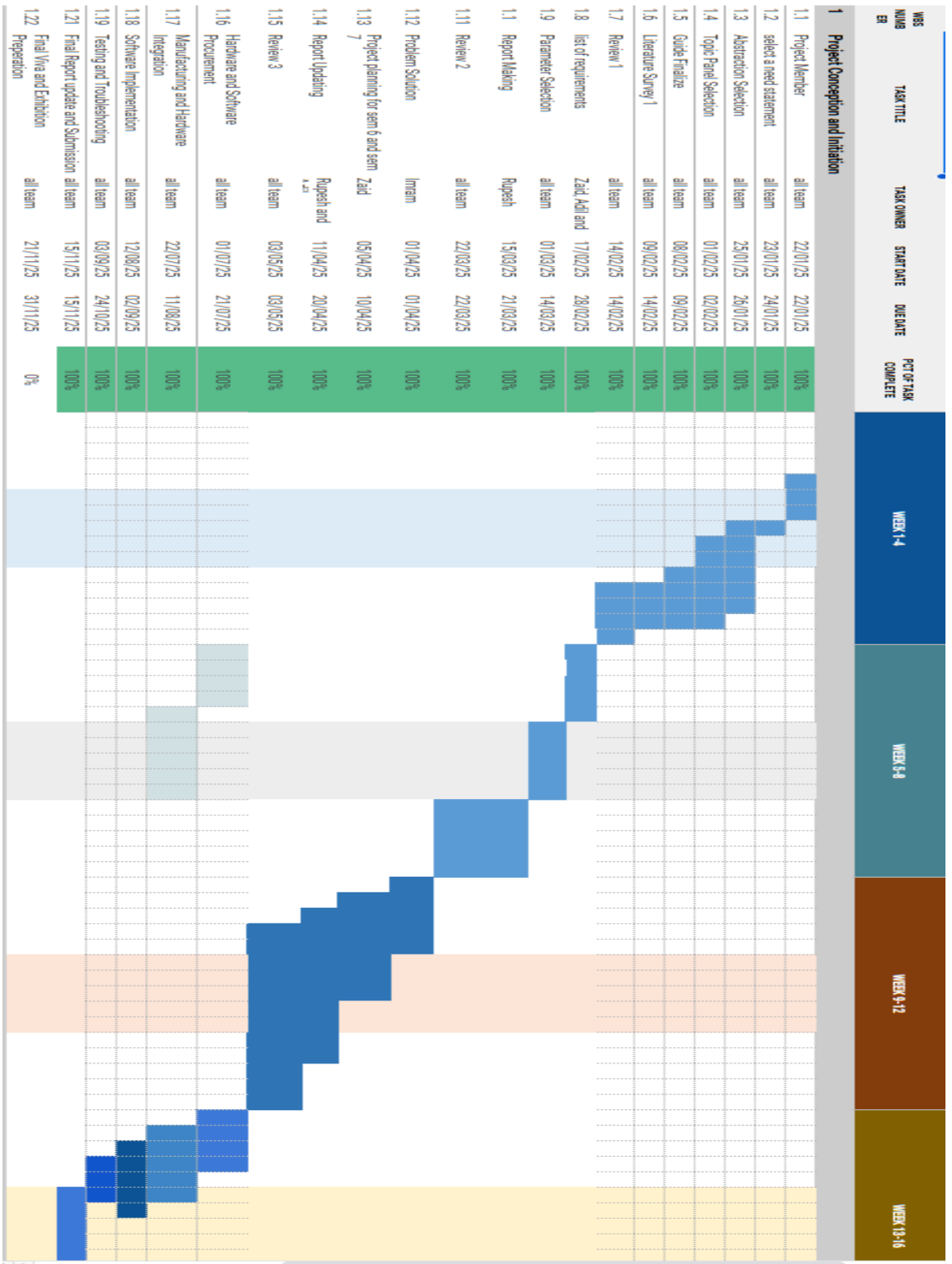
1.6.2 Block diagram of IOT based cotton quality monitoring system.

1.7 Pin Connection Diagram:



1.7.1 Pin Connection Diagram of Cotton Quality Monitoring System

1.8 Gantt Chart



1.8.1 Gantt Chart

1.9 List of Parameters

Sr. No.	Parameters	Description	Ideal Range	Measurement Unit	Sensor or Device	Cost Rupees																		
1.	Staple Length	The staple length is the average length of a group of Fibers that make up cotton yarn	<table><tr><th>Description</th><th>Staple Length (mm)</th><th>Staple Length (inches)</th></tr><tr><td>Short length</td><td><20.6</td><td><26/32</td></tr><tr><td>Medium length</td><td>20.6-25.4</td><td>26/32-32/32</td></tr><tr><td>Medium long length</td><td>26.2-27.8</td><td>33/32-35/32</td></tr><tr><td>Long</td><td>28.6-33.3</td><td>36/32-42/32</td></tr><tr><td>Extra long</td><td>>34.9</td><td>>44/32</td></tr></table>	Description	Staple Length (mm)	Staple Length (inches)	Short length	<20.6	<26/32	Medium length	20.6-25.4	26/32-32/32	Medium long length	26.2-27.8	33/32-35/32	Long	28.6-33.3	36/32-42/32	Extra long	>34.9	>44/32	Millimeters(mm) And Inches(inches).	Ultrasonic Sensor	100
Description	Staple Length (mm)	Staple Length (inches)																						
Short length	<20.6	<26/32																						
Medium length	20.6-25.4	26/32-32/32																						
Medium long length	26.2-27.8	33/32-35/32																						
Long	28.6-33.3	36/32-42/32																						
Extra long	>34.9	>44/32																						
2.	Moisture Content	Moisture content in cotton is crucial for various stages of cotton processing, from harvesting and ginning to storage and textile manufacturing,	according to information cited by the USDA, Ideal: 6–8% Acceptable: 8–10%	%percentage of the fiber's weight	Moisture Meter	500																		

3.	color Grade	Cotton color grading is a system used to classify the color of cotton fibres based on their appearance. The color of cotton can vary due to factors such as maturity, cleanliness, and processing techniques	<table><tr><td></td><td>White</td><td>Light Spotted</td><td>Spotted</td><td>Tinged</td><td>Yellow Stained</td></tr><tr><td>Good Middling</td><td>11-1</td><td>12</td><td>13</td><td>*</td><td></td></tr><tr><td>Strict Middling</td><td>21-2</td><td>22</td><td>23</td><td>24</td><td>35</td></tr><tr><td>Middling</td><td>31-3</td><td>32</td><td>33</td><td>34</td><td>35</td></tr><tr><td>Strict Low Middling</td><td>41-4</td><td>42</td><td>43</td><td>44</td><td>*</td></tr><tr><td>Low Middling</td><td>51-5</td><td>52</td><td>53</td><td>54</td><td>*</td></tr><tr><td>Strict Good Ordinary</td><td>61-6</td><td>62</td><td>63</td><td>*</td><td>*</td></tr><tr><td>Good Ordinary</td><td>71-7</td><td>*</td><td></td><td>*</td><td></td></tr><tr><td>Below Grade</td><td>81</td><td>82</td><td>83</td><td>84</td><td>85</td></tr></table>		White	Light Spotted	Spotted	Tinged	Yellow Stained	Good Middling	11-1	12	13	*		Strict Middling	21-2	22	23	24	35	Middling	31-3	32	33	34	35	Strict Low Middling	41-4	42	43	44	*	Low Middling	51-5	52	53	54	*	Strict Good Ordinary	61-6	62	63	*	*	Good Ordinary	71-7	*		*		Below Grade	81	82	83	84	85	1 White 2 Light spotted 3 Spotted 4 Tinged 5 Yellow Stained	Color Sensor (TCS3200)	400
	White	Light Spotted	Spotted	Tinged	Yellow Stained																																																							
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4.	R.D. Value	refers to the reflectance value, indicating how bright or dull a sample is, with higher values signifying whiter cotton. Having More Whiteness	General RD Value = 50 to 85 Good Variety of Cotton RD Value = 74 To 76	+b= Yellow ness RD = Brightn ess	RGB Sensor (TCS3200)	500																																																						

5.	Trash Content	Trash in cotton refers to non-fiber particles such as leaf, seedcoat, bark, grass, dust and other foreign matters.	The surface of the cotton sample is scanned by a video camera and the percentage of the surface area occupied by trash particles is calculated	Not Applicable (NA)	ESP Cam32	1500
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Table 1.9.1 List of parameters for cotton monitoring system

Sr. No	Parameters		Min Criteria		Max Criteria		Ideal Range	Real	Consider (Measurement)																																																						
1.	Staple Length (average length of a group of fibers)	1)Medium Length 2)Long Length	(mm) 1) >20.5 2) >29.5	(Inches) 0.807 1.161	(mm) <24.5 <30.5	(Inches) 0.964 1.200	>20 To <32.0mm	>20 To 32.5 mm & above	1) Staple Length Is Measured Using Device Ultrasonic Sensor.																																																						
2.	R D Value (Reflectance Value)	Brightnes s and Dullness Of Cotton.	General 50%	Good Variety 74%	General 85%	Good Variety 76 %	50 to 85%	74 to 76%	2) R D Value Is Calculated Using RGB)(TCS3200) Sensor.																																																						
3.	Moisture Content	Measures The Moisture Content Present In The Cotton.	6%		8%		6 to 8 %	8 to 10% (Acceptable)	3)Moisture Is Measured Using Moisture Sensor. 4) Color Grading Is Done By Color Sensor(TCS3200).																																																						
Description																																																															
4.	Trash Content	refers to non-fiber particles.	The surface of the cotton sample is scanned by a video camera and the percentage of the surface area occupied by trash particles is calculated				NA	NA																																																							
5.	Color Grade	used to classify the color of cotton fibers based on their appearance.	<table><tr><th></th><th>White</th><th>Light Spotted</th><th>Spotted</th><th>Tinged</th><th>Yellow Stained</th></tr><tr><td>Good Middling</td><td>11 - 1</td><td>12</td><td>13</td><td>-</td><td></td></tr><tr><td>Strict Middling</td><td>21 - 2</td><td>22</td><td>23</td><td>24</td><td>35</td></tr><tr><td>Middling</td><td>31 - 3</td><td>32</td><td>33</td><td>34</td><td>35</td></tr><tr><td>Strict Low Middling</td><td>41 - 4</td><td>42</td><td>43</td><td>44</td><td>-</td></tr><tr><td>Low Middling</td><td>51 - 5</td><td>52</td><td>53</td><td>54</td><td>-</td></tr><tr><td>Strict Good Ordinary</td><td>61 - 6</td><td>62</td><td>63</td><td>-</td><td>-</td></tr><tr><td>Good Ordinary</td><td>71 - 7</td><td>-</td><td></td><td>-</td><td></td></tr><tr><td>Below Grade</td><td>81</td><td>82</td><td>83</td><td>84</td><td>85</td></tr></table> 1. White 2. Light spotted 3. Spotted 4. Tinged 5. Yellow Stained					White	Light Spotted	Spotted	Tinged	Yellow Stained	Good Middling	11 - 1	12	13	-		Strict Middling	21 - 2	22	23	24	35	Middling	31 - 3	32	33	34	35	Strict Low Middling	41 - 4	42	43	44	-	Low Middling	51 - 5	52	53	54	-	Strict Good Ordinary	61 - 6	62	63	-	-	Good Ordinary	71 - 7	-		-		Below Grade	81	82	83	84	85	NA	NA	
	White	Light Spotted	Spotted	Tinged	Yellow Stained																																																										
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Good Ordinary	71 - 7	-		-																																																											
Below Grade	81	82	83	84	85																																																										

Table 1.9.2 Parameters and their criteria

	THE COTTON CORPORATION OF INDIA LTD. (A Government of India Undertaking) 'Kapas Bhavan', Plot No.3-A, Sector 10, CBD-Belapur, Navi Mumbai 400 614 Tel No: +91-22-27579217, Fax No: +91-22- 27576030, 27579219
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No.CCI/HO/PUR/2024

5th August 2024

GM/DGM/AGM/ Sr. Mgr./BM

The CCI Ltd,

Ahmedabad/Akola/Aurangabad/Adilabad/Bhatinda/Bhilwara/Coimbatore/Guntur

Hubli/ Indore/Kolkata/Rajkot/Rayagada/Sriganganagar/Sirsa/Warangal/ Mahabubangar

Sub : Minimum Support Price for Cotton Season : 2024-25

Dear Sir,

Please find enclosed herewith Order No.1/20/2024-25/MSP/Cotton/38-91 dated 29th July 2024 of Textile Commissioner, Ministry of Textiles, Government of India informing that the Government of India has fixed the support price of two basic varieties of cotton viz. Medium Staple Length Cotton having Staple Length of 24.5 to 25.5mm with micronaire value of 4.3 to 5.1 and Long Staple Length Cotton having Staple Length of 29.5 to 30.5mm with micronaire value of 3.5 to 4.3 of new crop of Seed Cotton (kapas) of MSP parameters for Cotton Season 2024-25 (October-September). The support price for Medium Staple Length cotton has been fixed at Rs.7121/- per quintal and that for Long Staple Length Cotton of above classes has been fixed at Rs.7521/- per quintal.

Based on the support prices for the above two basic classes of Seed Cotton and taking into account the quality differential, normal price differential and other relevant factors, the support prices for other classes of Seed Cotton (kapas) of MSP parameters for the Cotton Season 2024-25 (October-September) has been fixed by the Textile Commissioner in the aforesaid order alongwith various incentives/deductions for micronaire value and moisture content.

The above order of Textile Commissioner is self-explanatory and exhaustive, therefore the branches may effect MSP purchases as per detailed guidelines given in the above order.

The above MSP rates and deductions to be made from MSP rates may suitably be displayed in market yards/centre offices in local language for the information and benefit of cotton farmers.

All Branches should scrupulously follow the above guidelines while conducting the MSP purchases if any, during Cotton Season 2024-25.

Thanking you,

Yours sincerely,


 (Pranjal P. Joshi)

GENERAL MANAGER (PURCHASE)

Cotton Corporation of India LTD.

THE COTTON CORPORATION OF INDIA LTD.
Head Office: CBD Belapur, Navi Mumbai

Kind attention of all Cotton Buyers

We are pleased to inform our esteemed cotton buyers that CCI's platinum quality cotton of Crop Year 2022-23 is having following parameters.

S. No.	Parameter	Quality Parameters	
1	Staple length/UHML	30 mm+	29 mm+
2	Micronaire value	3.7 to 4.3	3.7 to 4.3
3	RD value	77+	77+
4	Fibre strength	30.5(±1.5) g/tex	29.5(±1.5) g/tex
5	Uniformity index	84%	83%
6	Trash	2%	2%
7	Moisture content	8%	8%

The date of commencement and other details of e-auction will be updated on the corporation's website <https://www.cotcorp.org.in>

SR.NO.	CLASSES OF COTTON	FIBRE QUALITY PARAMETERS		MINIMUM SUPPORT PRICE (MSP) FOR 2024-25	(₹ per quintal) NAMES OF THE INDICATIVE VARIETIES USED BY THE TRADE
		BASIC STAPLE LENGTH (2.5% SPAN LENGTH) IN MM	MICRONAIRE VALUE		
(I)	(II)	(III)	(IV)	(V)	(VI)
	Short Staple (20 mm & below)				
1		-	7.0 - 8.0	6621	Assam Comilla
2		-	6.8 - 7.2	6621	Bengal Deshi
	Medium Staple (20.5 mm – 24.5 mm)				
3		21.5 - 22.5	4.8 - 5.8	6871	Jayadhar
4		21.5 - 23.5	4.2 - 6.0	6921	V-797/ G.Cot.13/ G.Cot.21
5		23.5 - 24.5	3.4 - 5.5	6971	AK/Y-1(Mah & M.P.)/ MCU-7(TN)/SVPR-2 (TN)/PCO-2 (AP & Kar) / K-11(TN)
	Medium Long Staple (25.0 mm – 27.0 mm)				
6		24.5 - 25.5	4.3 - 5.1	7121	J-34(Raj)
7		26.0 - 26.5	3.4 - 4.9	7221	LRA-5166/KC-2(TN)
8		26.5 - 27.0	3.8 - 4.8	7271	F-414/H-777/J-34 Hybrid
	Long Staple (27.5 mm – 32.0 mm)				
9		27.5 - 28.5	4.0 - 4.8	7421	F-414/H-777/J-34 Hybrid
10		27.5 - 28.5	3.5 - 4.7	7421	H-4/H-6/MECH/RCH-2
11		27.5 - 29.0	3.6 - 4.8	7471	Shankar-6/10
12		29.5 - 30.5	3.5 - 4.3	7521	Bunny/Brahma
	Extra Long Staple (32.5 mm & above)				
13		32.5 - 33.5	3.2 - 4.3	7721	MCU-5/Surabhi
14		34.0 - 36.0	3.0 - 3.5	7921	DCH-32
15		37.0 - 39.0	3.2 - 3.6	8721	Suvin

Table 1.9.3 Parameters and their quality

2. LITERATURE SURVEY

Sr. No.	Research Paper Name	Publish Year	Description	Hardware and Software Requirements
1.	Cotton Appearance Grade Classification Based on Machine Learning. (2019 International Conference on Identification, Information and Knowledge in the Internet of Things (IIKI2019)).	Procedia Computer Science, Volume 174, in the year 2020	This Paper Presents a Method for Automatic Cotton Classification for different appearance grades. Based on a comprehensive comparison, our method performs better in the classification of cotton appearance grades. It utilizes PCANet, VGGNet, and Gabor filters for feature extraction, combined with SVM for classification. The method achieves high accuracy, especially when incorporating impurity identification. The study highlights the potential for automation in cotton grading, improving efficiency and reducing human error This paper searched for a feature extraction method for automatic cotton grade classification with high accuracy	Hardware: - 1. Industrial camera. 2. Light box with D65 light sources for image Acquisition
2.	Design and Experiment of a Sowing	Journal <i>Agriculture</i>	This paper presents a sowing quality monitoring system for cotton precision hill-drop	Hardware: - 1. Incremental

	Quality Monitoring System of Cotton Precision Hill-Drop Planters	Volume12 , Issue 8, Article 1117 in July 29, 2022	planters, designed to improve the intelligence and efficiency of cotton sowing operations. The system integrates a missed seedling monitoring module and a visualization module, utilizing color-coded electric eye sensors and LabVIEW software for real-time data processing and display. Bench and field tests demonstrated high accuracy, with sowing monitoring accuracy above 93% and missed seeding detection above 91%. The system addresses challenges in cotton sowing quality monitoring, particularly in detecting missed seeds and optimizing sowing parameters	rotary encoder. 2. color-coded electric eye color fiber optic sensor. 3. color code sensor amplifier. 4. industrial computing Shielded cable. Software: - LabVIEW for graphical programming and data visualization
3.	Quantitation of Fiber Quality and the Cotton Production-Processing Interface: A Physiologist's Perspective. by Judith M. Bradow and	published in 2000 in the <i>Journal of Cotton Science</i> , Volume 4, Issue 1.	Cotton quality is really important for both cotton farmers and the people who process it. Since the 1980s, the USDA has been helping by classifying cotton to make sure both sides are on the same page. But to keep things moving fast, sometimes they've had to make some	NA

	Gayle H. Davidonis		compromises, which can cause confusion. Now, new technology is being developed to measure cotton quality more accurately, reduce the differences in quality, and help both farmers and processors work better together	
4.	Study on the measurement and evaluation of cotton color using image analysis. Chong Heng, Lijun Chen, Hua Shen and Fumei Wang Published 17 July 2020	published on July 17, 2020 , In <i>Materials Research Express</i> , Volume 7, Article 075101	A fast and low-cost digital technique was used to test cotton color with high accuracy. The proposed LDL test method mainly contains LED lamps, digital CCD camera and correction equations caused by light source. Images captured by the experimental device contained the color data, from which the color parameters were obtained by image analysis	Hardware: 1. LED lamp 2.CCD Camera Software: 1. Image acquiring 2. Image Processing 3. Result Displaying
5.	Design and test of intelligent inspection and monitoring system for cotton bale storage based on RFID	published on March 16, 2022 , in the journal <i>Scientific Reports</i> , Volume 12, Article number: 4491.	This paper presents an intelligent inspection and monitoring system for cotton bale storage using RFID technology, wireless temperature and humidity monitoring, and handheld terminals. The system integrates RFID Positioning, temperature/humidity sensors, and a PSO-ANN algorithm for accurate data processing and positioning. It demonstrates	NA

			improved efficiency in cotton bale management, with positioning errors below 6.7% and temperature /Humidity monitoring errors under 8%. The hardware includes RFID tags, network/wireless card readers, temperature/humidity sensors, and handheld terminals, while the software utilizes Java for the handheld terminal system	
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Table 2.1 Literature Survey

3. PROBLEM DEFINITION

3.1 Challenges

1. Inaccurate and Time-Consuming Manual Monitoring: Manual measurements of cotton parameters are labor- intensive, prone to human error, and often lead to inconsistent data.
2. Inconsistent Quality Control: Without continuous and accurate data, it's difficult to ensure uniform quality, which impacts the cotton market grade and processing outcome.
3. Limited Data Access and Analysis: Existing systems do not offer automated data collection and analysis, making it difficult to track trends or make data-driven decisions to optimize cotton production.

3.2 Objectives

To develop an IoT-based automatic cotton monitoring system that integrates sensors. The system will measure:

1. Staple Length: The length of the cotton fibers, influencing their processing potential.
2. R.D. Value (Relative Density): The density of cotton fibers, which impacts on its quality and grade.
3. Moisture Content: The water content in the cotton fibers, which must be monitored to prevent mold and ensure proper storage.
4. Color Grade: The cotton's color, used to determine its grade and market value.
5. Trash Content: The amount of non-cotton material mixed with the cotton, affecting its purity and processing efficiency

3.3 Expected Impact

1. Enhanced Cotton Quality: Improved monitoring leads to better quality control and higher-grade cotton.
2. Operational Efficiency: Automation reduces the need for manual labor, minimizes errors, and accelerates the monitoring process.
3. Cost Reduction: By optimizing processes, the system helps reduce waste, improve yield, and minimize losses.

4. SYSTEM DEVELOPMENT

The proposed system integrates IoT sensors, a Raspberry Pi microcontroller, and image processing algorithms to monitor and evaluate cotton quality.

1. Staple Length: - The Parameter is Calculated Using Ultrasonic Sensor by measuring the distance in form of length in millimeter(mm).

2. Trash Content: - Capture a clear image of the cotton sample using a camera under uniform lighting. Preprocess the image by converting it to grayscale and applying thresholding to separate trash particles from clean fibers. Use `cv2.findContours()` to detect and isolate trash particles.

3. R D Value: -

The Reflectance Degree (R D) Value Is Calculate Using the RGB (Red, Green, Blue) Sensor (TCS3200). It Reads the Value For R, G, B From a Cotton Sample and by using The Formula We Calculate the R D Value.

Brightness Formula (Luminance/Y):

$$Y = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$

Normalize to Percentage Scale (0–255 mapped to 0–100%)

$$RD (\%) = \left(\frac{Y}{255} \right) \times 100$$

4.Color Grading: - The Color Grading Is Done by Using the TCS3200 Color Sensor captures red, green, and blue intensity values from the cotton sample. Based on predefined RGB thresholds, the system classifies cotton as White, Light Spotted, Spotted, Tinged etc.

5.Moisture Meter: - Measures the moisture content in cotton Sample by Using the Moisture Meter. The first sensor measures **resistance or capacitance** of the cotton sample, which changes with moisture levels and sensor reading when the cotton is completely dry and then sensor reading when the cotton is fully saturated with moisture (close to 100%).

Expected Impact

Improved Cotton Quality: Enhanced quality control through automated, objective assessment.

Operational Efficiency: Significant reduction in labor, time, and human error.

Cost Optimization: Lower losses due to quality degradation, resulting in higher profitability

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirement

Sr.no	Component	Description
1.	ESP8266	Acts as the central microcontroller to collect and process sensor data for wireless data transmission to the cloud.
2.	ESP Cam32	Captures images of cotton for Trash Content Detection.
3.	Moisture Meter	Measures the moisture content in cotton to avoid over/under drying.
4.	Ultrasonic Sensor	Use To Measure the Staple Length of Cotton sample.
5.	TCS3200 Color Sensor	Used for Color grading of cotton.
6.	RGB Sensor (TCS3200).	Calculates R.D. (Reflectance-Degree) value for brightness assessment.
7.	Industrial-grade Power Supply	To power all components reliably in field conditions.
8.	LCD Display Module	For local display of data and alerts.
9.	Trash Content Detection Module (Camera + Image Processing)	To detect non-fiber particles in the cotton.

Table 5.1 Hardware Requirements

Software Requirements

Sr.no	Software	Purpose
1.	Arduino IDE	Operating system for ESP 32, ESP Cam32
2.	Cloud Platform (Thing speak)	For storing and visualizing sensor data.
3.	Web Dashboard / Mobile App (Optional)	For remote monitoring and user interface.
4.	Machine Learning Libraries (scikit-learn, TensorFlow Lite)	For classification and prediction models (if implemented).
5.	Database (SQLite, MySQL etc.) (Optional)	For storing historical data.

Table 5.2 Software Requirements

6. TECHNICAL ASPECTS

Sr.no	Aspects	Specifications
1.	Sensors and Data Acquisition	1. Moisture Meter: Measures the moisture content in cotton, which affects storage and processing. 2. Color Sensor (TCS3200): - Detects the Color of cotton, useful for grading quality. 3.Esp32 Camera: - Detect Trash in Cotton Sample. 4.Ultrasonic Sensor: - Use to Measure the Staple Length of Cotton sample.
2.	Microcontroller/Embedded System	ESP8266. Collects sensor data, processes initial logic, and transmits data.
3.	Connectivity	Wi-Fi / GSM / LoRa / Zigbee: Used to transmit data to a central server or cloud. (Optional). Selection depends on range, data rate, and power consumption.
4.	Cloud Platform / Data Storage	Platforms like Thing Speak, IoT, Google Firebase, or Azure IoT Hub to Stores historical data.
5.	Data Processing and Analytics	Software (Python, Node.js, etc.) processes data to assess cotton quality. Algorithms for detecting anomalies or trends in quality metrics. Quality thresholds and alerts can be defined.

6.	User Interface	Web dashboard, mobile app, Think Speak dashboard for farmers, manufacturers, or quality controllers Visualizes sensor data, alerts, and quality reports
7.	Power Supply	Battery Management System Include Battery-powered (for remote fields) or standard AC power in ginning units.

Table 6.1: Technical Aspects

7. SENSORS DETAILS

1. Ultrasonic Sensor :

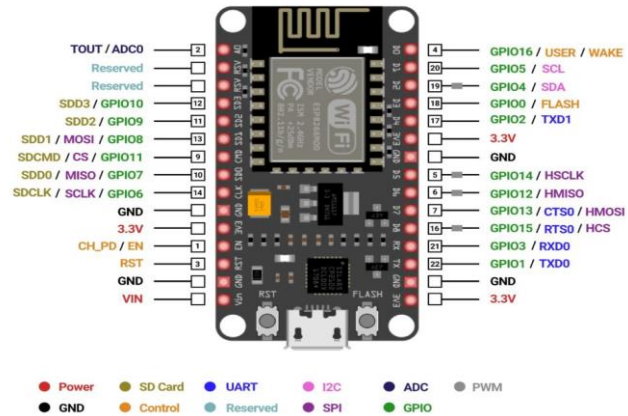
Model: - HC-SR04



Working Principle	Specifications	Function in Project	Pin Connection
Emits ultrasonic sound waves and measures time of flight for echo to return. Calculating distance as The distance is mapped to the average fiber length $\text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2}$ (staple length) of cotton.	• Power Supply: +5V DC • Quiescent Current: <2mA • Working current: 15mA • Effectual Angle: <15° • Ranging Distance: 2-400 cm • Resolution: 0.3 cm • Measuring Angle: 30° • Trigger Input Pulse width: 10uS • Dimension: 45mm x 20mm x 15mm • Weight: approx. 10 g	It is Used to Measures the Staple length, a key cotton sample parameter	In Esp8266 1. VCC: 3.3V shared with ESP8266, LCD, and Moisture sensor 2. TRIG: D8 3. ECHO: D7 4. GND: Common ground with ESP8266, LCD, and Moisture sensor

Table7.1: - Detail Description of Ultrasonic Sensor

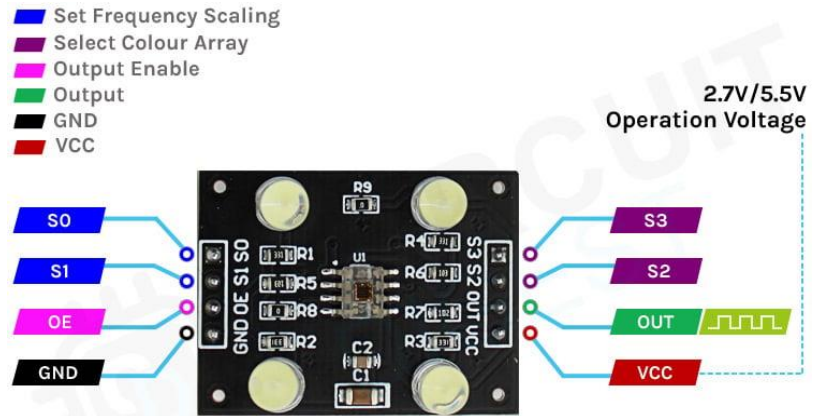
2. ESP8266:



Working Principle	Specifications	Function in Project	Pin Connection
1. Collects and processes sensor data. 2. Captures images for trash detection. 3. Connects to cloud (Thing Speak) for remote monitoring. 4. Acts as the IoT gateway for all components	1. Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106 2. Operating Voltage: 3.3V 3. Input Voltage: 7-12V 4. Digital I/O Pins (DIO): 16 5. Analog Input Pins (ADC): 1 6. UARTs: 1 7. SPIs: 1 8. I2Cs: 1 9. Flash Memory: 4 MB 10. SRAM: 64 KB 11. Clock Speed: 80 MHz 12. USB-TTL based on CP2102 is included onboard, Enabling Plug n Play 13. PCB Antenna 14. Small Sized module to fit smartly inside your IoT projects	The ESP32 acts as the central microcontroller. It reads data from sensors (Moisture, Color, RGB, etc.), processes it, and transmits readings to the cloud platform (Thing Speak). The ESP-CAM captures images of the cotton to analyze trash	1. 3V3: 3.3V rail 2. GND: Common ground 3. D0: TCS3200 S0 4. D1: LCD SDA 5. D2: LCD SCL 6. D3: TCS3200 S1 7. D4: TCS3200 S2 8. D5: TCS3200 S3 9. D6: TCS3200 OUT 10. D7: Ultrasonic ECHO 11. D8: Ultrasonic TRIG 12. A0: Soil Moisture Sensor A0 13. VIN: ESP32-CAM VCC

Table7.2: - Detail Description of ESP8266

3. TCS3200:



Working Principle	Specifications	Function in Project	Pin Connection
1. Measures the intensity of red, green, and blue light reflected by the cotton sample. Calculates Reflectance. 2. Degree (R.D.) using luminance (brightness): $Y = 0.299R + 0.587G + 0.114B$ The normalized brightness. 3. value represents whiteness, higher R.D. indicates better cotton quality.	1. Operating voltage range of 7V to 5.5V. 3. Operating temperature range of -40°C to 85°C 4. Output frequency range of 0KHz to 2.7KHz 5. High sensitivity to a wide range of colors. 6. Can be programmed for different color detection ranges (red, blue, green, clear)	1. Evaluates brightness and cleanliness of cotton. 2. Helps classify the grade based on reflectance.	In ESP8266 VCC: 3.3V GND: Common ground S0: D0 S1: D3 S2: D4 S3: D5 OUT: D6

Table7.3: - Detail Description of TCS3200

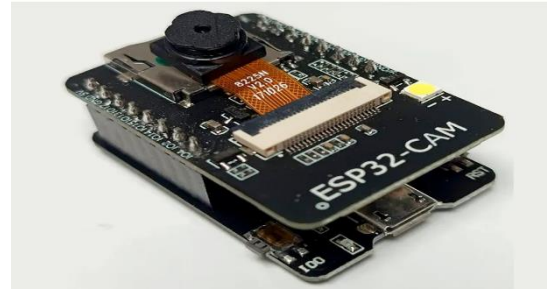
4. Moisture Sensor with Analog to Digital Converter (ADC):



Working Principle	Specifications	Function in Project	Pin Connection
Works on resistance or capacitance change due to water content in cotton fibers. Higher moisture → higher conductivity (lower resistance). The analog signal is converted to Digital in form of percentage.	Type: Capacitive or resistive moisture sensor Operating Voltage: 3.3V–5V Output: Analog voltage proportional to moisture level Measurement Range: 0–100% relative moisture	1. Monitors moisture content (ideal 6–8%). 2. Prevents over-drying or under-drying of cotton.	In ESP8266 VCC: 3.3V GND: - A0: A0 pin

Table7.4: - Detail Description of Moisture with ADC

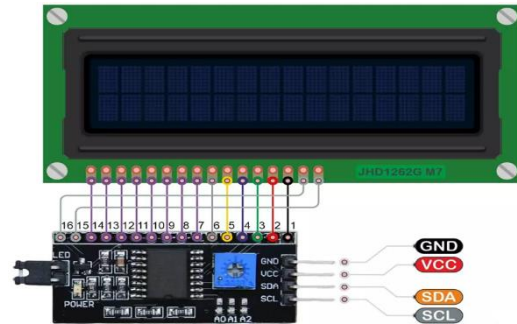
5. ESP32 CAM:



Working Principle	Specifications	Function in Project	Pin Connection
The OV2640 camera captures the image. ESP32 processes and compresses it (JPEG). The data is then streamed via Wi-Fi or stored on microSD.	1. Built-in OV2640 camera 2. ESP32-S Wi-Fi + Bluetooth chip 3. 4MB PSRAM, microSD card support 4. Image resolution up to 1600×1200 5. Operates at 3.3V	Captures top-view images of cotton under uniform light. Image is processed using OpenCV to detect non-fiber particles. Enhances the purity evaluation of cotton samples	In Esp8266 1. VCC: - Connected to 3V3. 2. GND: - Common GND

Table7.5: - Detail Description of ESP32 CAM

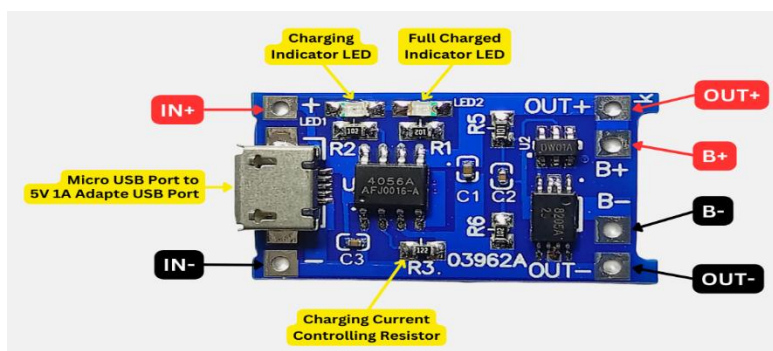
6. 16x2 LCD Display with I2C Module:



Working Principle	Specifications	Function in Project	Pin Connection
Microcontroller sends data through I2C (SDA/SCL). PCF8574 converts I2C signals to parallel signals. HD44780 controller displays characters on the 16×2 screen.	1. 16 characters × 2 lines 2. HD44780 controller 3. I2C interface (PCF8574). 4. Uses 4 pins: VCC, GND, SDA, SCL 5. Works on 5V, low power, LED backlight	1. Displays real-time data: moisture, color grade, R.D. value, etc. 2. Gives local feedback even without internet.	In Esp8266 VCC: Connected to the 3.3V power rail. GND: Connected common ground. SDA: pin D1. SCL: pin D2.

Table7.6: - Detail Description Of 16x2 LCD Display with I2C Module

7. TP4056:

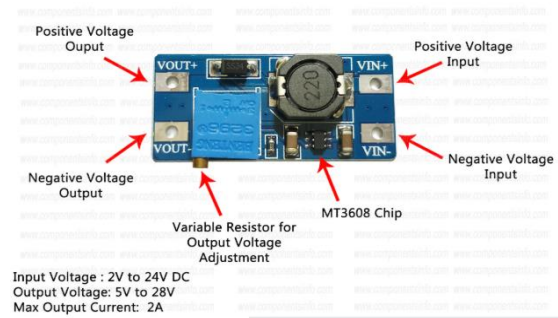


Working Principle	Specifications	Function in Project	Pin Connection
1. Takes 5V input from USB. 2. TP4056A charges the battery and stops at 4.2V. 3. Controls and limits charging current. 4. LEDs show charging (red) and full (blue). 5. DW01A + FS8205A protect the cell from overcharge, over-discharge, and short circuit.	1. Type: 18650 Charger & Current Protection Board 2.Module IC: TP4056 SOP-8 Chip 3. Cell Type: Li-Ion, Li-Pol 4. Cell Voltage: 3.7 V 5.PowerConnector: Micro-B USB 6. Input Voltage: 4V to 8V 7. Input Current (Max.): 1050 mA (~1A) 8. Charging Voltage: 4.2 V 9. Charging Current: 1 Amp (default) 10. Charging Precision: 1.5%	It is a part of the Battery Management System used to charge the lithium cell, allowing the project to operate without an external power supply.	1. OUT+: To battery VCC 2. OUT-: To battery GND 3. B+: To rocker switch input 4. B-: To boost converter VOUT-

Table7.7: - Detail Description of TP4056

8. MT3608 DC To DC Step Up

Boost Converter Module:



Working Principle	Specifications	Function in Project	Pin Connection
1. Takes a low DC input voltage. 2. MT3608 IC switches the inductor rapidly to store and release energy. 3. This boosts the voltage to a higher level, controlled by the feedback potentiometer. 4. Provides stable, adjustable output for powering higher-voltage loads.	1. Input: 2V–24V DC 2. Output: 5V–28V DC (adjustable) 3. Output current: up to 2A 4. Efficiency: up to 93% 5. Soft-start feature 6. Very low quiescent current (~200 μ A) 8. Overheat protection 9. Compact, low-cost module 10. Uses MT3608 high-efficiency boost IC	It is a part of the power management system used to boost the battery voltage, ensuring the project receives a stable and higher operating voltage.	1. VIN+: ESP8266 3.3V 2. VIN-: ESP8266 GND 3. VOUT+: From rocker switch output 4. VOUT-: To TP4056 B-

Table7.8: - Detail Description of MT3608 DC To DC Step Up Boost Converter Module

9. 3.7V Battery and Rocker Switch:



Parameter	3.7V Lithium Battery	Rocker Switch
Working Principle	Stores electrical energy and supplies constant DC power to the circuit.	Manually opens or closes the power line to turn the circuit ON or OFF.
Specifications	• Nominal Voltage: 3.7V • Fully Charged: 4.2V • Rechargeable Li-ion/Li-Po • High energy density	• SPST ON/OFF type • Mechanical switch • Suitable for low-voltage DC loads
Function in Project	Powers the system for standalone operation without external supply.	Controls the delivery of battery power to the project.
Pin Connection	VCC: TP4056 OUT+ GND: TP4056 OUT-	Input: TP4056 B+ Output: Boost converter VOUT+

Table7.9: - Detail Description Of 3.7V Battery and Rocker Switch.

10. Cloud Integration:



Function	Description	Role in Project
Wi-Fi Connection	Connects to Wi-Fi using SSID and password.	Enables internet access for cloud upload.
ThingSpeak Communication	Sends data using channel ID and API key.	Allows remote monitoring and data logging.
Ultrasonic Sensor	Measures distance and calculate staple length (mm).	Determines cotton staple length for quality analysis.
Moisture Meter	Reads soil moisture and maps it to percentage.	Monitors moisture level of cotton samples/field.
TCS3200 Color Sensor	Captures RGB values and identifies cotton color grade.	Evaluates cotton quality based on color intensity.
LCD Display	Shows moisture, staple length, RGB values, and grade in rotating pages.	Provides real-time local feedback to the user.
Data Transmission	Sends all sensor data to Thing Speak every 10 seconds.	Ensures continuous cloud-based tracking.
Setup () Function	Initializes Wi-Fi, sensors, LCD, and communication modules.	Prepares the system for operation.
Loop () Function	Continuously reads, processes, displays, and uploads data.	Maintains Data monitoring and updates.

Table7.10: - Detail Description of Cloud Integration

8. CIRCUIT DOCUMENTATION

This circuit is designed to integrate various sensors and modules to monitor environmental conditions and communicate data wirelessly. It includes an ESP8266 microcontroller for processing and communication, a TP4056 for battery management, a step-up boost converter for voltage regulation, and several sensors including an ultrasonic sensor, a soil moisture sensor, and a TCS3200 color sensor. The circuit also features an LCD display for local data visualization and an ESP32-CAM for potential image capabilities.

Component List

1. Ultrasonic Sensor (HC-SR04)

Description: Used for measuring distance by using ultrasonic waves.

Pins: VCC, TRIG, ECHO, GND

2. ESP32-CAM

Description: A microcontroller with camera capabilities for image capture and processing.

Pins: 5V, GND, IO12, IO13, IO15, IO14, IO2, IO4, VOT, VOR, VCC, IO0, IO16, 3V3, Antenna

3. Step Up Boost Power Converter, Adjustable Voltage Regulator

Description: Boosts the voltage from the battery to a higher level required by the circuit.

Pins: VOUT+, VOUT-, VIN+, VIN-

4. Rocker Switch

Description: Used to control the power supply to the circuit.

Pins: Output, Input

5. TCS3200 Color Sensor

Description: Detects color by measuring the intensity of light in the red, green, and blue spectrum.

Pins: GND, OUT, S2, S3, VCC, LED, S0, S1

6. Battery 3.7V

Description: Provides power to the circuit.

Pins: VCC, GND

7. LCD 16x2 with I2C Interface

Description: Displays data such as sensor readings and status messages.

Pins: GND, VCC, SDA, SCL

8. ESP8266

Description: A microcontroller used for processing and WiFi communication.

Pins: D0, D1, D2, D3, D4, 3V3, GND, D5, D6, D7, D8, RX, TX, A0, RSV, SD3, SD2, SD1, CMD, SDO, CLK, EN, RST, VIN

9. TP4056

Description: A lithium battery charger module.

Pins: OUT-, B-, B+, OUT+, IN-, IN+

10. Moisture Meter

Description: Measures the moisture content in the soil.

Pins: VCC, GND, D0, A0

9. WIRING DETAILS

1. Ultrasonic Sensor (HC-SR04)

- **VCC:** Connected to 3.3V power rail shared with ESP8266, LCD, and Soil Moisture Sensor.
- **TRIG:** Connected to ESP8266 pin D8.
- **ECHO:** Connected to ESP8266 pin D7.
- **GND:** Connected to common ground shared with ESP8266, LCD, and Soil Moisture Sensor.

2. ESP32-CAM

- **VCC:** Connected to the 3.3V power rail.
- **3V3:** Connected to the common ground. Step Up Boost Power Converter
- **VIN+:** Connected to ESP8266 3.3V pin.
- **VIN-:** Connected to ESP8266 GND pin.
- **VOUT+:** Connected to the output of the rocker switch.
- **VOUT-:** Connected to TP4056 B- pin.

3. Rocker Switch

- **Input:** Connected to TP4056 B+ pin.
- **Output:** Connected to Step Up Boost Power Converter VOUT+ pin.

4. TCS3200 Color Sensor

- **VCC:** Connected to the 3.3V power rail.
- **GND:** Connected to common ground.
- **S0:** Connected to ESP8266 pin D0.
- **S1:** Connected to ESP8266 pin D3.
- **S2:** Connected to ESP8266 pin D4.
- **S3:** Connected to ESP8266 pin D5.
- **OUT:** Connected to ESP8266 pin D6.

5. Battery 3.7V

- **VCC:** Connected to TP4056 OUT+ pin.
- **GND:** Connected to TP4056 OUT- pin.

6. LCD 16x2 with I2C Interface

- **VCC:** Connected to the 3.3V power rail.
- **GND:** Connected to common ground.
- **SDA:** Connected to ESP8266 pin D1.
- **SCL:** Connected to ESP8266 pin D2.

7. ESP8266

- **3V3:** Connected to the 3.3V power rail.
- **GND:** Connected to common ground.
- **D0:** Connected to TCS3200 S0 pin.
- **D1:** Connected to LCD SDA pin.
- **D2:** Connected to LCD SCL pin.
- **D3:** Connected to TCS3200 S1 pin.
- **D4:** Connected to TCS3200 S2 pin.
- **D5:** Connected to TCS3200 S3 pin.
- **D6:** Connected to TCS3200 OUT pin.
- **D7:** Connected to Ultrasonic ECHO pin.
- **D8:** Connected to Ultrasonic TRIG pin.
- **A0:** Connected to Soil Moisture Sensor A0 pin.
- **VIN:** Connected to ESP32-CAM VCC pin.

8. TP4056

- **OUT+:** Connected to Battery VCC pin.
- **OUT-:** Connected to Battery GND pin.
- **B+:** Connected to Rocker Switch input pin.
- **B-:** Connected to Step Up Boost Power Converter VOUT- pin.

9. Moisture Meter

- **VCC:** Connected to the 3.3V power rail.
- **GND:** Connected to common ground.
- **A0:** Connected to ESP8266 A0 pin.

10. DOCUMENTED CODE

The code for the ESP8266 microcontroller is responsible for reading sensor data, displaying it on the LCD, and sending it to Thing Speak for remote monitoring. Below is a summary of the key functionalities:

- **Wi-Fi Connection:** The ESP8266 connects to a Wi-Fi network using the provided SSID and password.
- **ThingSpeak Communication:** Data is sent to ThingSpeak using the provided channel ID and API key.
- **Ultrasonic Sensor:** Measures distance and calculates the staple length in millimeters.
- **Moisture Meter:** Reads the moisture level and maps it to a percentage.
- **TCS3200 Color Sensor:** Measures RGB values and determines the cotton grade based on color intensity.
- **LCD Display:** Cycles through pages displaying soil moisture, staple length, and RGB values with the cotton grade.
- **Data Transmission:** Sends sensor data to Thing Speak every 10 seconds. The code is structured to initialize the hardware in the setup () function and continuously read and process sensor data in the loop () function. The LCD displays the data in a user-friendly format, and the Thing Speak platform is used for remote data logging and analysis.

11. SYSTEM FLOW CHART

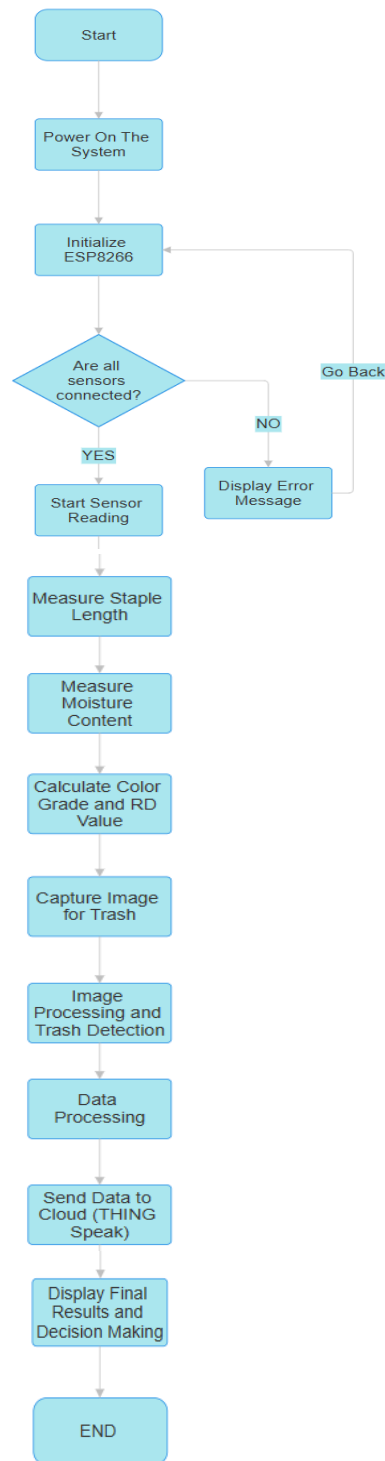


Fig :11.1: Working Flow of System

12. PROJECT PLANNING FOR SEM VI AND SEM VII

Semester VI: Project Report Phase Duration: *22 January 2025 – 30 June 2025*

Objective: Prepare a comprehensive project report including title, abstract, objectives, methodology, literature review, and implementation planning.

Phase 1: Topic Finalization & Synopsis (Week 1–2) Tasks:

1. Finalize project topic and title.
2. Define project domain (e.g., IoT, Automation, Embedded Systems, etc.).
3. Submit synopsis with team members and obtain guide approval.

Deliverables:

1. Approved project title.
2. Synopsis signed by guide.

Phase 2: Literature Review & Abstract (Week 3–5) Tasks:

1. Study 5–7 relevant research papers or existing projects.
2. Draft an abstract summarizing the project scope, problem statement, and goals.
3. Identify and highlight the gap your project addresses.

Deliverables:

1. Literature Review (formatted in IEEE style).
2. Project Abstract.

Phase 3: Feasibility Study & Planning (Week 6–7) Tasks:

1. Conduct feasibility analysis in terms of cost, time, and technology.
2. Identify preliminary hardware and software requirements.
3. Define major milestones for Semester VII.

Deliverables:

1. Feasibility Report.
2. Gantt Chart or Timeline for Semester VII.

Phase 4: Methodology & Block Diagram (Week 8–10) Tasks:

1. Describe step-by-step working of the proposed system.
2. Create block diagrams and flowcharts.
3. List system components and their interactions.

Deliverables:

1. Project Methodology Chapter.
2. System Block Diagram.
3. Data Flow Diagram / Flowchart.

Phase 5: Chapter Writing (Week 11–14) Tasks:

Write the following chapters:

- Introduction
- Literature Review
- Proposed System
- Hardware & Software Requirements
- Methodology
- Expected Output
- References
- Ensure adherence to college formatting guidelines.

Deliverables:

- Complete Draft Report.

Phase 6: Internal Review & Final Submission (Week 15–20) Tasks:

1. Get the report reviewed by your guide and make necessary revisions.
2. Prepare a mini-PowerPoint for internal oral review.
3. Finalize and bind the report.

Deliverables:

1. Soft copy of report and presentation.
2. Final hardbound report.

Semester VII: Implementation & Completion Phase

Duration: *1 July 2025 – 30 November 2025*

Objective: Convert the project report into a fully implemented hardware-software system with working results and demonstration.

Phase 1: Hardware & Software Procurement (Week 1–3) Tasks:

1. Purchase or order necessary hardware components (e.g., sensors, MCUs).
2. Set up all development tools (IDEs, libraries, etc.).

Deliverables:

1. Component list and purchase report (with bill photos).
2. Software setup report.

Phase 2: Manufacturing & Hardware Integration (Week 4–6) Tasks:

1. Assemble the system design.
2. Create physical housing (chassis, body, or PCB) if needed.
3. Connect all components for testing.

Deliverables:

1. System Design.
2. Photos of assembly stages.
3. Physical setup documentation.

Phase 3: Software Implementation (Week 7–9) Tasks:

1. Develop and upload firmware or application code.
2. Conduct unit testing for all software modules.
3. Integrate software with the hardware setup.

Deliverables:

1. Source code files.
2. Screenshots of code execution and outputs.
3. Software flow explanation.

Phase 4: Testing & Troubleshooting (Week 10–11) Tasks:

1. Perform full system testing.
2. Fix identified hardware or software issues.
3. Document all changes made during debugging.

Deliverables:

1. Testing report.
2. Photos/videos of test results

Phase 5: Final Output & Result Recording (Week 12–13) Tasks:

1. Demonstrate full working of the system.
2. Record and analyze system output and performance data.
3. Capture final setup images.

Deliverables:

1. Demonstration video.
2. Final system photographs.
3. Output result charts/logs.

Phase 6: Final Report Update & Submission (Week 14–16) Tasks:

1. Update the report with final results, images, and source code.
2. Get final approval from the project guide.
3. Submit hardbound report.

Deliverables:

1. Complete final report with all proof of work.
2. Final presentation slides.

Phase 7: Final Viva & Exhibition Preparation (Last 2 Weeks) Tasks:

1. Prepare the final presentation (PPT) explaining system flow, implementation, and results.
2. Rehearse for viva-voce and project demonstration.
3. Participate in any exhibition or competitions, if applicable.

Deliverables:

1. Viva-ready PPT.

2. Working system for demonstration.
3. Participation/Exhibition certificate (if any).

13. CONCLUSIONS AND FUTURE SCOPE

Conclusion: -

The IoT-Based Cotton Quality Detecting System revolutionizes the traditional cotton quality assessment process by integrating sensor technology, cloud computing, and IoT. This system ensures significantly improving efficiency in cotton farming and processing.

Key Takeaways:

1. **Automation & Efficiency:** Reduces manual labor and speeds up the assessment process, leading to cost savings.
2. **Data-Driven Insights:** Provides farmers and processors with actionable insights to optimize operations.
3. **Scalability & Future Enhancements:** The system can be extended with AI-driven predictive analytics, blockchain for traceability, and automation in cotton sorting.

Future Scope:

1. **AI & Machine Learning** – Automated defect detection, grading, and quality prediction for faster processing. **Advanced Imaging & Computer Vision** – High-resolution cameras and scanners for precise quality analysis.
2. **Blockchain for Transparency** – Tracks cotton quality from farm to textile factories to prevent fraud and ensure authenticity.
3. **Automation & Robotics** – Reduces human errors in quality testing, making the process faster and more reliable.
4. **Sustainable & Eco-Friendly Testing** – Non-destructive and chemical-free testing methods to ensure sustainability.
5. **Cloud-Based Data Storage & Analysis** – Helps farmers and textile industries access real-time quality reports for better decision-making.

14. REFERENCES

Publications:-

1) Chong Heng, Lijun Chen, Hua Shen, Fumei Wang, “Study on the measurement and evaluation of cotton color using image analysis”, 2020, PP 78-87.

Ref(https://www.researchgate.net/publication/342793709_Study_on_the_measurement_and_evaluation_of_cotton_color_using_image_analysis)

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Ref(https://www.researchgate.net/publication/279626743_Quantitation_of_Fiber_Quality_and_the_Cotton_Production-Processing_Interface_A_Physiologist's_Perspective).

3) Yan Lv, Ying Gao, Eric Rigall, Lin Qi, Feng Gao and Junyu, “Cotton Appearance Grade Classification Based on Machine Learning”, ‘ScienceDirect’, 2020, PP 729- 734.

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Ref(<https://www.mdpi.com/2077-0472/12/8/1117>)

5) Weipeng Zhang, Bo Zhao, Qizhi Yang, “Design and test of intelligent inspection and monitoring system for cotton bale storage based on RFID”, ‘Scientific report’, 2022, PP 1-16.

Ref(https://www.researchgate.net/publication/367951824_Design_and_Test_of_Intelligent_Inspection_and_Monitoring_System_for_Cotton_Bale_Storage_based_on_RFID).

Web references:-

1) Textile Study Center

Ref(<https://textilestudycenter.com/length-parameters-cotton-fibers/>)

2) Importance of RD Value || Assess the Cotton Rates ||

Sahoo Textile Academy Ref(https://www.youtube.com/watch?v=_5APQy7flw)

3) Cotton trash measurement using image analysis

Ref(<https://www.cotton.org/beltwide/proceedings/getPDF.cfm?year=1996&paper=WDD003.pdf>)

4) Cotton Color Grading and

Classification Ref(<https://textilesbar.com/cotton-color-grading-and-classification/>)

5) Cotton Works By Cotton Incorporated. Ref(<https://cottonworks.com/en/topics/sourcing-manufacturing/fiber-science/cotton-fiber-qualities-and-evaluation/>)

6) Ministry Of Textiles Cotton Sector Ref(<https://www.texmin.nic.in/sector-industry/cotton>)

7) The Classification Of Cotton

Ref(https://www.depts.ttu.edu/aaec/icac/archive/pdf/Markets_and_Marketing_optimized/Grading/The_Classification_of_Cotton-Agricultural_Handbook_566.pdf)

15. APPENDIX

Appendix A: System Architecture Diagrams

- A.1: Diagram of the IoT-based system architecture (including sensors, microcontroller, and communication modules).
- A.2: Data flow diagram of the cotton monitoring system.
- A.3: Detailed block diagram showing the interaction between various system components (e.g., sensors, server, cloud storage).

Appendix B: Hardware Specifications

- B.1: Detailed specifications for sensors used (e.g., moisture sensor, trash content sensor, step length sensor).
- B.2: Microcontroller specifications (e.g., Arduino, Raspberry Pi, etc.).
- B.3: Communication protocol details (e.g., Wi-Fi, LoRa, Bluetooth) used for data transmission.

Appendix C: Software Code

- C.1: Source code for sensor data collection and processing (e.g., Arduino code, Python scripts).
- C.2: Code for data transmission to cloud or database.
- C.3: Algorithms used for processing cotton parameters (e.g., step length calculation, moisture detection algorithm).

Appendix D: Data Collection Forms and Survey Questionnaires

- D.1: Survey used to collect farmer or user feedback on cotton quality monitoring.
- D.2: Data collection form used for field testing the IoT system (if applicable).

Appendix E: Raw Data and Results

- E.1: Sample raw data collected from sensors (e.g., moisture content, cotton color, step length).
- E.2: Data logs showing the system performance and cotton quality metrics over time.
- E.3: Statistical analysis or charts showing the relationship between cotton quality parameters.

Appendix F: Test Cases and Testing Results

F.1: Test cases used to validate the system's performance.

F.2: Results of system testing (e.g., sensor accuracy, data transmission reliability).

F.3: Performance comparison of the system with traditional cotton monitoring methods.

Appendix G: User Manuals and Documentation

G.1: User manual for operating the IoT-based cotton monitoring system.

G.2: Instructions for setting up and configuring the sensors and microcontroller.

G.3: Troubleshooting guide for system issues.

Appendix H: References to Literature and Additional Resources

H.1: Relevant research papers and articles that support the system's design.

H.2: Technical documentation on IoT protocols, sensor specifications, or cloud-based systems us